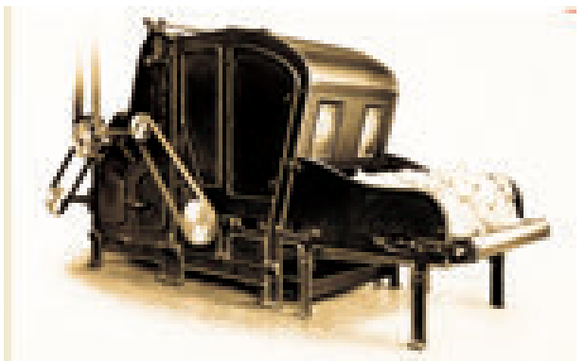
Cotton is removed mechanically from the seed balls by the cotton gin. The ginned cotton is then pressed into bales and sent to the factories to be spun into yarn.

## i. Manufacturing process

The main processes are bale breaking and cleaning, carding, combing, spinning, weaving, scouring, bleaching and dyeing.

a) **Bale breaking and cleaning:** The tightly pressed cotton fibres from the bales are loosened in a machine and the impurities falls out. Another machine removes more impurities until sheets of loose fibre like cotton wool emerge ready for carding.



Bale breaking

b) **Carding:** The fibres emerge from the carding rolls as a thin filmy sheet which passes between two rolls and is formed is to a thick rope called "**Slivers**".



Carding

c) **Combing:** This process removes more short hairs and makes fibres more parallel.



Combing

d) **Spinning:** The combed sliver is now converted into yarn by spinning. Scouring and Bleaching is usually done after weaving to enable the cloth to be dyed easily.



Spinning

## ii. Properties:

| | | |
|---|---|---|
| Lustre | : | Low |
| Tenacity/ Strength | : | Medium |
| Elastic Recovery | : | Low |
| Elongation | : | 7% |
| Resilience | : | Low |
| Density | : | 1.54 g/cm$^3$ |